

ASSEMBLY OF THE LOCAL TYPE II EXCLUSION THEOREM FOR NAVIER–STOKES BLOWUP

GUILLEM CORDERO

ABSTRACT. We assemble the local Type II exclusion argument for the three-dimensional incompressible Navier–Stokes equations in standard PDE language. The argument starts from a suitable weak solution near a putative singular point and uses the local Caffarelli–Kohn–Nirenberg concentration theorem, a local Type I tangent criterion, the local single-core Type II criterion, the localized scale-translation representation, the local multibubble and cascade exclusion, the compact-cylinder Caccioppoli regularity theorem, and the scale-collapse dichotomy. Every positive local Type II concentration is exhausted by the single-core, multibubble/cascade, rough-core, or scale-collapse alternatives, and each alternative is excluded by the corresponding theorem. Consequently no Type II singularity occurs at the point.

1. INTRODUCTION

Let (u, p) be a suitable weak solution of the three-dimensional incompressible Navier–Stokes equations

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \quad \nabla \cdot u = 0,$$

in a neighborhood of a possible singular point $z_0 = (x_0, T)$. This paper assembles the local Type II part of the analysis from the local concentration reduction together with the local Type II components stated below.

The local formulation starts from parabolic rescalings. For $r > 0$, set

$$Q_r(z_0) = B_r(x_0) \times (T - r^2, T),$$

and define the pressure-normalized Caffarelli–Kohn–Nirenberg quantities

$$C(z_0, r) = r^{-2} \int_{Q_r(z_0)} |u|^3 dx dt,$$

$$D(z_0, r) = r^{-2} \int_{T-r^2}^T \int_{B_r(x_0)} |p(x, t) - (p)_{B_r(x_0)}(t)|^{3/2} dx dt.$$

The pressure is always understood modulo functions of time, as in the local energy inequality and the CKN theory [CKN82].

The local concentration theorem says that a singular point has positive local CKN concentration. After the local Type I tangent criterion removes nonzero bounded parabolic tangent limits, the remaining positive concentration is the local Type II regime. The local Type II analysis divides that regime into the following PDE alternatives.

- (i) A single selected core with bounded compact-cylinder suitable package and nondegenerate localized scale-translation gauges.
- (ii) A multibubble, cascade, or gauge-degenerate configuration.
- (iii) Failure of local windowed H^1 control in a bounded renormalized core.
- (iv) Collapse of the renormalized scale.

The single-core alternative is excluded by the local compact criterion. The multibubble and cascade alternatives are excluded by the local decoupling and scale-rigidity theorems. The rough-core alternative is excluded by the compact-cylinder Caccioppoli theorem. The scale-collapse alternative is excluded either by a finite-cost condition controlled by the scale-drift cost or by reduction to a stationary or generalized self-similar profile in a parameter regime covered by the relevant Liouville theorem.

Theorem 1.1 (Local Type II exclusion, assembled form). *Let (u, p) be a suitable weak solution near (x_0, T) . If (x_0, T) is in the local Type II regime of [Theorem 2.2](#), then (x_0, T) is not singular.*

Here the Type II regime means that the Type I tangent alternative is absent. The remaining local alternatives are exhausted and excluded by the five local Type II components.

2. LOCAL CONCENTRATION AND TYPE I–TYPE II SPLIT

Theorem 2.1 (Positive local concentration at singular points). *If (x_0, T) is singular, then*

$$\limsup_{r \downarrow 0} (C((x_0, T), r) + D((x_0, T), r)) > 0.$$

Equivalently, there are $\eta > 0$ and $r_n \downarrow 0$ such that

$$C((x_0, T), r_n) + D((x_0, T), r_n) \geq \eta \quad \text{for all } n.$$

Proof. This is the contrapositive of CKN epsilon regularity. If the displayed limsup were zero, then for sufficiently small r the CKN quantity would be below the universal regularity threshold, and (x_0, T) would be regular. $\square$

Given such a sequence, define the parabolic rescalings

$$u_n(y, s) = r_n u(x_0 + r_n y, T + r_n^2 s), \quad p_n(y, s) = r_n^2 p(x_0 + r_n y, T + r_n^2 s).$$

Then (u_n, p_n) is suitable on every fixed backward cylinder once n is large enough, and the unit cylinder retains positive normalized CKN mass.

Definition 2.2 (Local Type II sequence). A positive local concentration sequence is called a local Type II sequence if no selected-window rescaling has a nonzero bounded parabolic tangent on a compact cylinder.

3. COMPACT-CYLINDER DICHOTOMY

For a suitable pair (v, q) on a backward cylinder, write

$$A_v(R) = R^{-1} \operatorname{ess\,sup}_{-R^2 < s < 0} \int_{B_R} |v(y, s)|^2 dy,$$

$$E_v(R) = R^{-1} \int_{-R^2}^0 \int_{B_R} |\nabla v|^2 dy ds,$$

and let $C_{v,q}(R)$, $D_{v,q}(R)$ denote the corresponding scaled local L^3 and pressure quantities.

Theorem 3.1 (Local compactness dichotomy). *After passing to a subsequence of the local rescalings (u_n, p_n) , exactly one of the following alternatives holds.*

(i) *For every fixed $R < \infty$,*

$$\sup_n (A_{u_n}(R) + E_{u_n}(R) + C_{u_n, p_n}(R) + D_{u_n, p_n}(R)) < \infty.$$

In this case a subsequence converges locally in the suitable weak class on compact backward cylinders.

- (ii) *For some fixed $R < \infty$, the preceding supremum is infinite. Then a bounded rescaled cylinder carries an additional local concentration, giving a multibubble or cascade alternative.*

Proof. The first alternative is precisely the compact-cylinder suitable package. If it holds, the standard compactness theorem for suitable weak solutions gives a subsequence with local weak suitable convergence, strong convergence in subcritical local spaces, and pressure convergence modulo spatial means. If it fails, then at least one component of the local suitable package is unbounded on a fixed cylinder. The local Caccioppoli inequality shows that boundedness of $C + D$ on a comparable larger cylinder controls $A + E$ on a smaller one. Thus failure of the compact-cylinder package forces additional bounded-cylinder concentration, which is the local multibubble or cascade alternative. $\square$

4. LOCAL PDE COMPONENTS

This section states the local PDE components used in the assembly theorem.

Theorem 4.1 (Local single-core criterion). *A local Type II sequence with one selected core, bounded compact-cylinder suitable package, nondegenerate localized scale-translation gauges, compact-ball pressure decomposition, local Caccioppoli bounds, and finite localized Type II cost cannot occur.*

Proof. Finite localized cost gives windows on which the localized dissipation tends to zero. Compact-cylinder bounds give convergence on the same windows. The low-dissipation limit is spatially constant on the selected core. If the constant is zero, the selected CKN mass vanishes, contradicting positive local concentration. If the constant is nonzero, the sequence has a nonzero bounded parabolic tangent, contradicting the definition of a local Type II sequence. $\square$

Theorem 4.2 (Localized scale-translation representation). *On a genuine single-core local Type II sequence, the localized scale and centering conditions can be imposed on compact cylinders whenever the localized Jacobian is nondegenerate. In the resulting variables (V, P) ,*

$$\partial_\tau V + (V \cdot \nabla)V + \nabla P = \nu \Delta V + a(\tau)(V + y \cdot \nabla V) + b(\tau) \cdot \nabla V, \quad \nabla \cdot V = 0,$$

and the pressure is determined modulo functions of time by the compact-ball pressure decomposition. Failure of the localized nondegeneracy condition is a multibubble or gauge-degenerate alternative.

Proof. The localized scale and centering functionals have an invertible derivative on the selected core. The quantitative inverse function theorem gives the local scale-translation choice on each compact selected window. Differentiating the rescaled variables gives the displayed equation, and the pressure transforms with the usual freedom to add functions of time. If the derivative is not invertible, a unique core has not been selected, which is precisely the local gauge-degenerate alternative. $\square$

Theorem 4.3 (Local multibubble and cascade exclusion). *No local multibubble, cascade, or gauge-degenerate Type II configuration occurs.*

Proof. If compact-cylinder bounds fail, or if active CKN mass splits, another active bounded-cylinder concentration core is selected. Comparable cores are combined or remain in the multibubble case. Strict same-point cascades are reduced to scale-rigidity or nonlinear no-splitting. Separated physical cores decouple in active local frames. The local decoupling and scale-rigidity theorems then reduce every such configuration either to the single-core criterion or to a nonzero bounded parabolic tangent, both of which are impossible in the local Type II regime. $\square$

Theorem 4.4 (Compact-cylinder Caccioppoli regularity). *For the renormalized Type II sequences arising from the local construction, the local pressure reconstruction modulo functions of time, the bounded critical local norm on the cylinders under consideration, and the bounded or window-integrable modulation coefficients give uniform local windowed H^1 control by the local Caccioppoli estimate. Consequently the rough-core alternative, meaning failure of this local windowed H^1 bound in a bounded renormalized core, cannot occur.*

Proof. The local energy inequality is pulled back through the absolutely continuous scale-translation chart. A compactly supported cutoff gives the local Caccioppoli estimate. The pressure term is estimated modulo constants by the compact-ball pressure decomposition and Calderon–Zygmund/harmonic bounds. The local L^2 and L^3 terms are controlled on bounded cylinders, and the modulation terms are controlled by the established coefficient bounds. This yields the windowed H^1 bound. $\square$

Theorem 4.5 (Scale-collapse exclusion alternatives). *Let (V, P, a, b, λ) be a renormalized Type II rescaling satisfying*

$$\partial_\tau \log \lambda = a, \quad \lambda(\tau) \rightarrow 0,$$

and suppose a localized L^2 core has a positive floor. Then

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau) d\tau = \infty.$$

The finite-cost scale-collapse rescaling is excluded by the scale-drift cost comparison. Alternatively, on thick scale-collapse windows with compactness, modulation convergence, and a stationary omega-limit, the rescaling produces a nonzero stationary or generalized self-similar profile, which is excluded by the corresponding Liouville theorem.

Proof. The identity $\partial_\tau \log \lambda = a$ implies

$$\int_{\tau_1}^{\tau_2} a(\tau) d\tau = \log \lambda(\tau_2) - \log \lambda(\tau_1).$$

If $\lambda(\tau) \rightarrow 0$, the right-hand side tends to $-\infty$. Hence $\int a_- = \infty$, and the positive localized L^2 floor gives divergence of $\int a_- M_R$. The cost conclusion follows from the finite-cost control theorem. The structural alternative follows by translating thick windows, using compactness and modulation convergence to pass to an autonomous reduced limit, and then applying the stationary omega-limit and Liouville theorem. $\square$

Theorem 4.6 (Local Type II alternative decomposition). *Every positive local Type II concentration sequence belongs to one of the following alternatives: compact single-core finite cost, multibubble/cascade or gauge degeneracy, rough-core loss of local windowed H^1 , or scale collapse.*

Proof. The compact-cylinder dichotomy gives either compact-cylinder suitable bounds or additional bounded-cylinder concentration. The latter is the multibubble/cascade alternative. Under compact-cylinder suitable bounds, localized scale-translation degeneracy is the gauge-degenerate alternative; the nondegenerate case gives the single-core renormalized variables. In this single-core case the selected-window compact argument either has finite localized Type II cost, loses local windowed H^1 control in a bounded renormalized core, or enters the scale-collapse alternative. These cases are the alternatives listed above. $\square$

5. ASSEMBLY THEOREM

Theorem 5.1 (Local Type II exclusion theorem). *Let (u, p) be a suitable weak Navier–Stokes solution near (x_0, T) . If (x_0, T) is in the local Type II regime of [Theorem 2.2](#), then (x_0, T) is not singular.*

Proof. Suppose, for contradiction, that (x_0, T) is a Type II singular point. By [Theorem 2.1](#), there is a sequence of parabolic rescalings with positive local CKN concentration. By the definition of the local Type II regime, no selected-window rescaling has a nonzero bounded parabolic tangent.

Apply the compact-cylinder dichotomy, [Theorem 3.1](#). If the compact-cylinder suitable package fails, then the sequence enters the local multibubble or cascade alternative, which is excluded by [Theorem 4.3](#). Thus we may assume compact-cylinder suitable bounds.

If the localized scale-translation conditions are degenerate, the sequence is again in the multibubble or gauge-degenerate alternative and is excluded by [Theorem 4.3](#). Otherwise [Theorem 4.2](#) gives the renormalized scale-translation variables and the local pressure representation on the selected single core.

For this renormalized single-core sequence, consider the possible failures of the selected-window compact argument. If the localized Type II cost is finite, then [Theorem 4.1](#) excludes the sequence. If local windowed H^1 control fails in a bounded renormalized core, [Theorem 4.4](#) excludes that alternative. If the renormalized scale collapses while a localized core remains nonzero, [Theorem 4.5](#) excludes the scale-collapse alternative.

By the local Type II alternative decomposition, [Theorem 4.6](#), these are all positive local Type II concentration cases. Each case has been excluded, contradicting the assumed Type II singularity. $\square$

Corollary 5.2 (Exclusion after Type I separation). *No singular point of a suitable weak solution belongs to the local Type II regime.*

Proof. This is [Theorem 5.1](#) applied at each putative singular point after the positive local concentration reduction. $\square$

6. CONCLUSION

The local concentration theorem reduces singularity formation to positive CKN concentration. The Type I tangent alternative is separated off by definition of the local Type II regime. The five local Type II theorems then exhaust the remaining possibilities: compact single-core finite cost, multibubble/cascade or gauge degeneracy, rough-core loss of local windowed H^1 , and scale collapse. Each possibility is excluded. Hence no local Type II singularity remains.

REFERENCES

- [CKN82] Luis Caffarelli, Robert Kohn, and Louis Nirenberg. Partial regularity of suitable weak solutions of the navier–stokes equations. *Communications on Pure and Applied Mathematics*, 35(6):771–831, 1982.